

Option D: Path difference = $1.60 \text{ nm} = 2.7 \lambda$, so incorrect.

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D

$$y = \frac{1}{2}at^2 \dots\dots\dots(1)$$

$$x = ut \dots\dots\dots(2)$$

from (1) & (2)

$$y = \frac{ax^2}{2u^2}, \text{ i.e. } y \propto x^2$$

$$p:q:r:s = x^2:(2x)^2:(3x)^2:(4x)^2 \\ = 1:4:9:16$$

D

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D

E